WTW 158 Worksheet 5 on Limit laws and Continuity

A. TEXTBOOK PROBLEMS
p 102 no 21, 23, 39, 45, 49
p196 no 39, 45, 50
p124 no 19, 23, 41, 45

B. MORE PROBLEMS

1. Given the function $f(x) = \begin{cases} e^{-x+1} & \text{if } x < 1 \\ k & \text{if } x = 1 \\ \frac{2}{x} & \text{if } x > 1 \end{cases}$

Find k (if possible) such that
(i) f is left continous in $x = 1$ (ii) f is right continuous in $x = 1$ (iii) f is continuous in $x = 1$

2. $f(x) = \begin{cases} \frac{\cos x - 1}{\sin x} & \text{if } x < 0 \\ 1 & \text{if } x = 0 \\ \frac{1}{\ln(x+1)} & \text{if } x > 0 \end{cases}$

Investigate the continuity of f in $x = 0$.
3. State the domain on which the following functions are continuous.
(i) $y = \arcsin x$ (ii) $f(x) = \frac{\tan x}{\sqrt{1-x^2}}$

C. TEST QUESTIONS
1. Find the limit (if it exists):
1.1 $\lim_{x\to 2} \sec \frac{\pi x}{3}$ 1.2 $\lim_{x\to 1} \frac{1-\sqrt{2x^2-1}}{x-1}$ 1.3 $\lim_{x\to 6^-} \frac{|x-6|}{x-6}$ 1.4 $\lim_{x\to 2} \frac{\sin(x^2-4)}{x-2}$
2. Consider the function:

$$f(x) = \begin{cases} \frac{\sin 2x}{x} & \textit{if } x \neq 0 \\ \ln a & \textit{if } x = 0 \end{cases}$$

For which value of a will f be continuous at $x = 0$?

(a)	$a = 0$	(b)	$a = 1$	(c)	$a = e^2$	(d)	$a = 2$	(e)	$a = e$
(f)	None of these								

D. YOUTUBE VIDEOS
Kahn Academy Continuity at a point.

E. SELECTED ANSWERS
A p 102 no 21. $\frac{1}{6}$ 23. $-\frac{1}{9}$ 39. Squeeze Theorem, 0 45. $-\infty$ 49 (a) (i) 5 (ii) -5 (b) No, does not exist
p 196 39. $\frac{5}{3}$ 45. $\frac{1}{2}$ 50. $\frac{1}{2}$ p 124 19. Does not exist 23. $f(2) = 3$ 41. f is discont in $x = -1$, f is cont in $x = 1$
45. $c = \frac{2}{3}$
B. 1. (i) $k = 1$ (ii) $k = 2$ (iii) f cont for no value of k
2. $\lim_{x\to 0^-} f(x) = 0, \lim_{x\to 0^+} f(x) = \infty$, so f is discont in $x = 0$ 3(i) $[-1,1]$ (ii) $(-1,1)$
C. 1.1 -2 1.2 -2 1.3 -1 1.4 4 2. (c)